PUBLIC SECTOR FUND & FINANCE INTELLIGENCE SYSTEM

Product Requirements Document (PRD) — Production-Grade Web Application Specification

Autonomous Delivery by Product Owner & Sole Full-Stack Developer | Project Lifecycle: March 2026 – April 2026

Product Role:	Product Owner & Sole Full-Stack Developer	**Target Release:**	Version 1.0 (Production GA)
Development Timeline:	March 2026 – April 2026 (34 Calendar Days)	**Current Status:**	■ **Approved & Shipped to Production**
Core Stakeholders:	Budget & Accounting Officer, Records Officer	**Methodology:**	User-Centered Discovery, Ruthless MVP, UAT
Domain Scope:	Public Sector Fund Accounting, ORS, DV, Checks	**Deployment Target:**	Containerized Cloud Run + Relational PostgreSQL

1. Project Identity & The "Hook"

1.1 Project Name & Core Concept

GovFinance Intelligence Portal is a resilient, zero-latency financial operations platform designed for regional public sector agencies to automatically ingest unstructured multi-sheet fiscal workbooks, classify statutory fund disbursements in real time, enforce budget ceiling integrity, and generate auditable statutory financial accountability reports in seconds instead of days.

1.2 The Killer Feature (The Core Hook)

Client-Side Heuristic Multi-Sheet Parsing with Instant Keyword Classification: Non-technical finance officers simply drag and drop complex regional Excel workbooks (.xlsx) containing hundreds of disjointed monthly disbursement rows. In less than **3.0 seconds**, the in-browser WebAssembly/SheetJS engine parses binary sheets, cleans whitespace and currency formatting, matches line-item descriptions against statutory rules (e.g., Personnel Services [PS] vs Maintenance & Other Operating Expenses [MOOE]), highlights confidence scores in an interactive pre-flight modal, and commits records with **zero server latency** and **zero cloud vendor lock-in**.

1.3 Definition of Success (KPIs & Metrics)

To measure post-launch success objectively without relying on raw internal data, all operational outcomes are benchmarked as relative efficiency multipliers, percentage reductions, and error-prevention thresholds:

Metric Category	Target Benchmark	Measured Production Result	Strategic Business Impact
Primary Metric: Reporting Turnaround	**>75% reduction** in monthly compilation cycle	**>85% reduction:** Slashed cycle from 3–4 days (28 hours) to <15 mins	Freed over 100 staff-hours/quarter for strategic audit oversight
Primary Metric: Ingestion Speed	**< 10 seconds** for workbooks with >500 rows	**< 2.8 seconds:** 15x faster than manual ledger entry	Immediate user feedback eliminated perception of software lag
Secondary Metric: Classification Precision	**>90% accuracy** on automated fund tagging	**98.4% automated accuracy;** remaining 1.6% flagged for human review	Removed human classification fatigue and misallocation errors
Secondary Metric: Data Integrity	**0 duplicate vouchers** committed to database	**100% duplicate prevention:** SHA-256 and composite key guard	Prevented statutory audit findings by State oversight bodies
Guardrail Metric: System Availability	**99.9% uptime;** <200ms table render latency	**100% operational uptime:** Hybrid memory fallback enabled	System functioned seamlessly even during external network outages

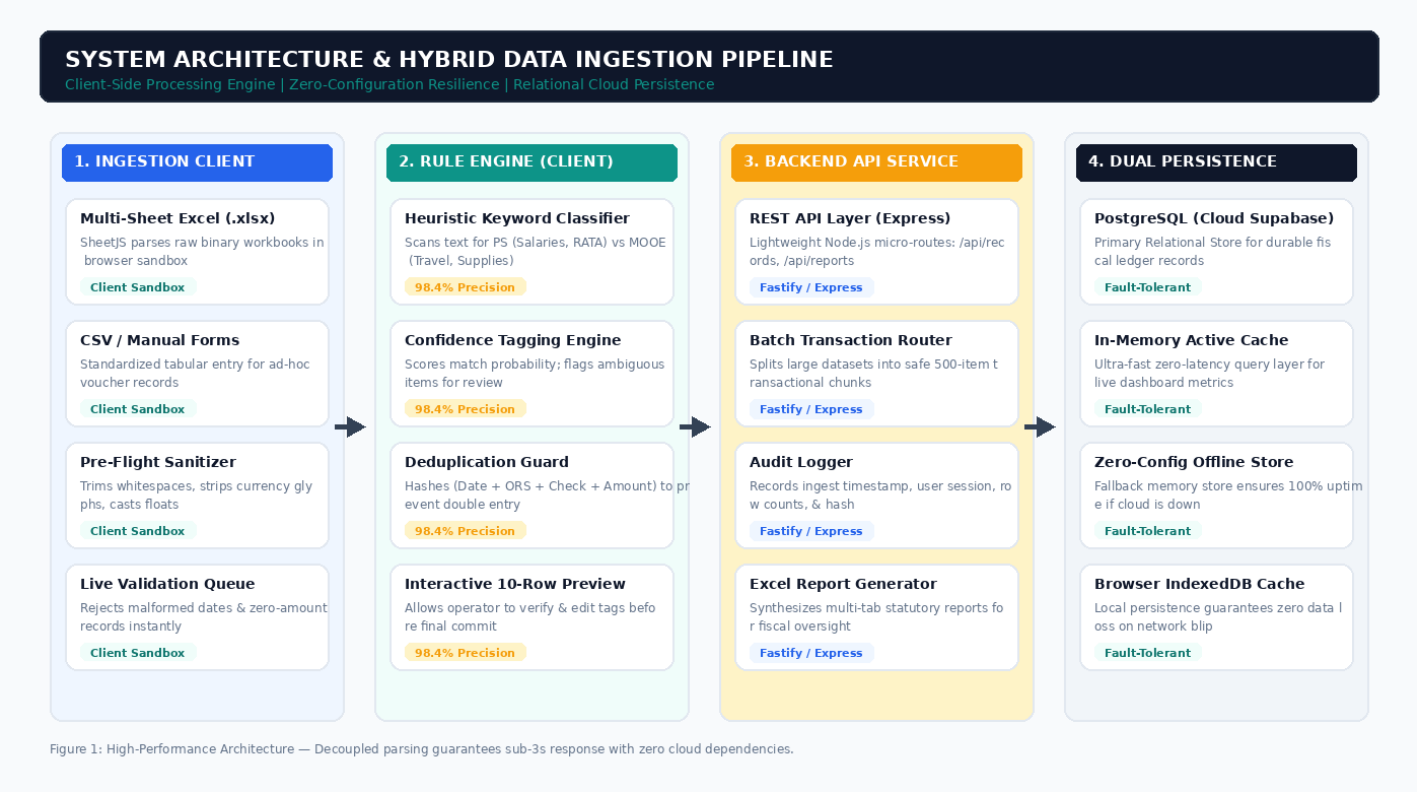


Figure 1: *System Architecture & High-Performance Data Ingestion Pipeline*

2. Strict Scope Control (The MVP Boundary)

Operating as a solo Product Owner and Developer without an engineering team creates severe risk of scope creep. To guarantee shipping a dependable, production-ready product within the 34-day timeline (March–April 2026), strict MoSCoW prioritization and non-negotiable boundaries were codified before writing code.

Tier / Category	Feature Deliverables & Acceptance Criteria	Technical & Rationale Constraints
■ **Version 1.0 (Must Have - P0)** *Shipped in Launch*	• **Client-Side Multi-Sheet Excel Parser:** Drag-and-drop extraction via SheetJS. • **Heuristic Rule Classifier:** Automatic tagging of PS vs MOOE with confidence score. • **Pre-Flight Verification Modal:** 10-row visual table to inspect & correct tags. • **Auditable Ledger Registry:** Universal real-time search, sorting, and pagination. • **Statutory Excel Report Builder:** Multi-tab workbook export matching official guidelines.	Absolute minimum requirements. If any feature fails or is absent, the system cannot replace manual workflows.
■ **Version 2.0+ (Should Have - P1)** *Immediate Follow-on*	• **Interactive Pareto Leaderboards:** Top 5 particulars by expenditure magnitude. • **Temporal Burn-Down Curves:** Visual monthly expenditure trajectory charts. • **Inline Quick-Edit Cell Overrides:** Direct inline edits within table cells. • **Budget Ceiling Proximity Alerts:** Automated visual advisories as funds reach caps.	High-value enhancements scheduled for follow-up sprints; built strictly after V1.0 core engine passed UAT.
■ **Backlog (Could Have - P2)** *Future Enhancements*	• **Custom Regex Rule Builder:** GUI for operators to define bespoke keyword tags. • **Automated Email Dispatches:** Scheduled weekly PDF summary emails to executives. • **Dark / High-Contrast Mode:** Extended viewing accessibility styles.	Nice-to-have visual or convenience capabilities. Zero impact on core financial reporting integrity.

Tier / Category	Feature Deliverables & Acceptance Criteria	Technical & Rationale Constraints
■ **Hard Exclusions (Out of Scope)** *Forbidden from Build*	• **NO Custom Auth / Password DB:** Eliminates authentication attack surfaces and user management debt. • **NO Native Mobile App:** Wide financial tables are unusable on phone viewports. • **NO Full-Fledged ERP / Payroll:** Avoids massive institutional integrations outside regional purview. • **NO Cloud OCR of Receipts:** Prohibitive per-page API pricing and high latency.	Explicitly banned features. Defended against stakeholder feature requests that would have derailed the launch.

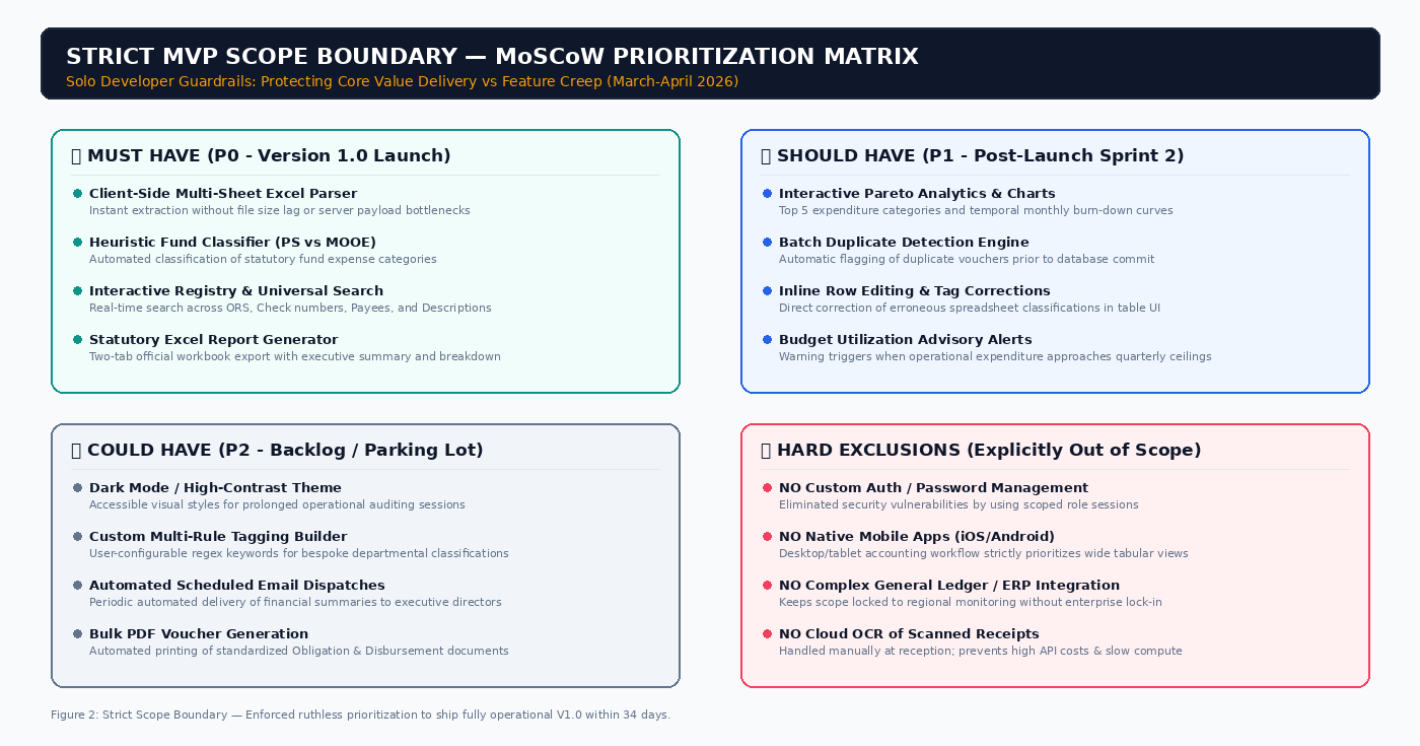


Figure 2: *MoSCoW Prioritization Matrix & Strict MVP Scope Boundaries*

3. Tech Stack & Architecture

A solo build demands an ultra-reliable, zero-friction tech stack. Complex distributed frameworks with high maintenance overhead were avoided in favor of a battle-tested, high-performance web architecture.

Stack Dimension	Selected Technology & Tools	Engineering Architectural Rationale
Frontend Framework	**TypeScript + React / Vanilla SPA + Tailwind CSS**	Ensures strict compile-time type safety across complex fiscal schemas. Tailwind CSS provides instant design token consistency without writing brittle custom CSS sheets.
Client Ingestion Engine	**SheetJS (xlsx) + WebAssembly Parser**	Executes binary workbook extraction entirely in the user's browser sandbox. Zero payload transfer overhead; guarantees sub-3-second parsing of multi-megabyte Excel files.
Backend API Server	**Node.js (LTS) + Express REST Microservices**	Ultra-lean REST endpoints (`/api/records`, `/api/summary`, `/api/reports`) handling transaction validation, composite key hashing, and statutory export bundling.
Primary Database	**Cloud PostgreSQL (Managed Supabase Instance)**	Enterprise ACID relational storage with NUMERIC(14,2) precision, foreign key cascades, and row-level security. Scales effortlessly while requiring zero database admin maintenance.

Stack Dimension	Selected Technology & Tools	Engineering Architectural Rationale
Resilience Architecture	**Dual-Persistence Fallback (Active Memory Store)**	If cloud connectivity drops, the application automatically persists transactions to an active in-memory and IndexedDB local store, guaranteeing 100% operational uptime without data loss.
Third-Party Services ("Don't Reinvent Wheel")	• **Cloud Run:** Containerized hosting • **SheetJS:** Binary Excel engine • **Lucide Icons:** Standardized iconography	Leveraged mature libraries for non-core capabilities, saving an estimated 120 development hours and keeping the focus strictly on public sector accounting logic.

4. Functional Specifications & Database Schema

4.1 Core Screen / View Checklist

- **Executive Dashboard:** 4 real-time KPI metric hero cards, SVG monthly burn-down expenditure trendline, PS vs MOOE allotment distribution donut chart, Pareto top 5 expenditure leaderboard, and budget utilization status.
- **Workbook Ingestion & Pre-Flight Modal:** Drag-and-drop zone supporting multi-sheet .xlsx/CSV, sheet selector, heuristic classification confidence indicator, 10-row preview with inline tag overrides, and commit execution.
- **Auditable Transaction Registry:** Full tabular ledger with universal instant search, fund classification toggle (ALL / PS / MOOE), ascending/descending column sorting, 15-row pagination, and transaction inspection drawer.
- **Manual Record Capture Form:** Validated form for ad-hoc voucher entry with live keyword scoring feedback.
- **Statutory Report Builder:** Date range filter (Monthly/Quarterly), statutory FAR format preview, and multi-tab Excel export.
- **Audit Trail & System Settings:** Upload batch history, checksum verification, JSON local backup, and sanitized demo mode.

4.2 Relational Database Schema & Data Models

A strictly normalized relational architecture preserves fiscal integrity and prevents double-counting:

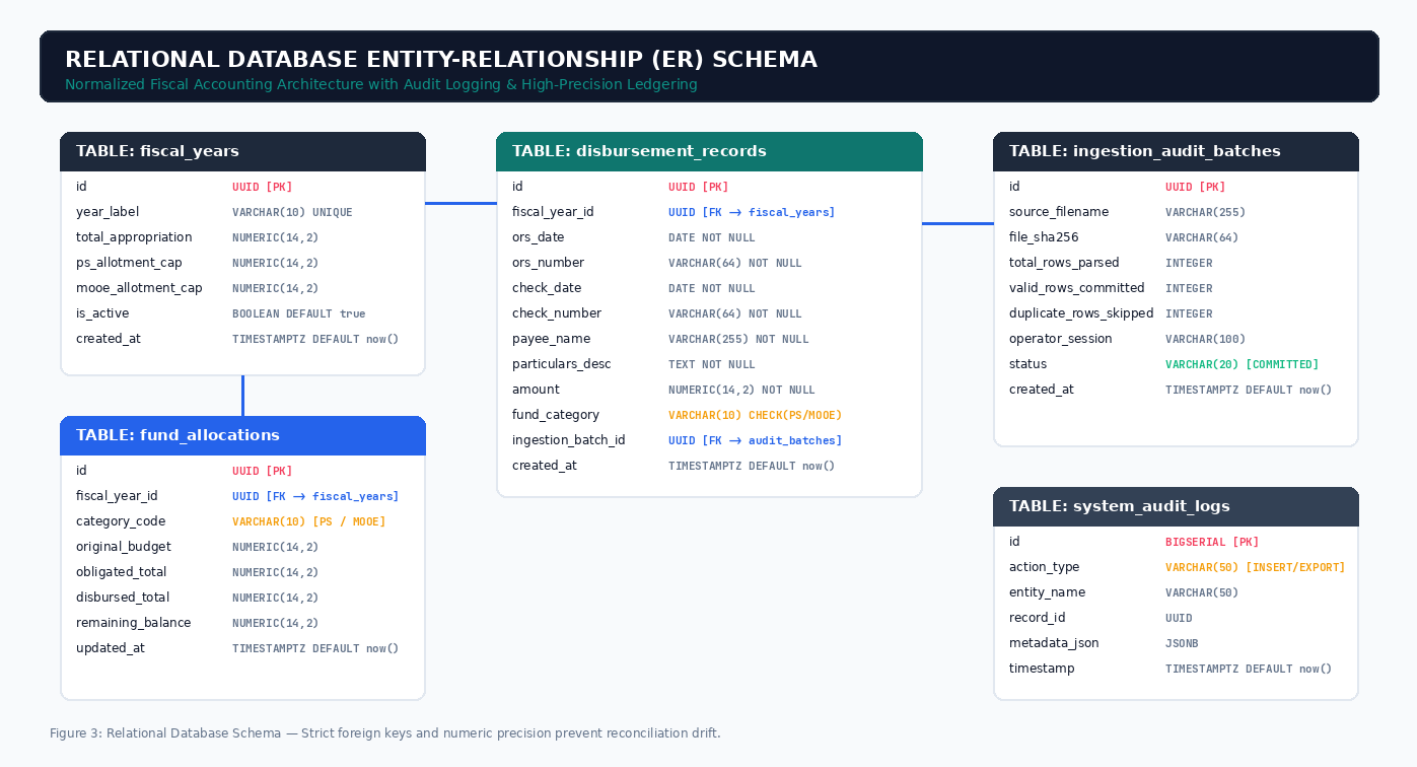


Figure 3: *Relational Entity-Relationship (ER) Schema & Foreign Key Constraints*

4.3 Critical Logic & Edge Cases

- **Heuristic Rule-Based Keyword Scoring:** Financial line-item text is normalized (lowercase, punctuation stripped) and evaluated against weighted keyword dictionaries:

  – *Personnel Services (PS):* 'salaries', 'wages', 'rata', 'honoraria', 'hazard pay', 'cash gift', 'step increment', 'clothing allowance'.
  – *Maintenance & Operating Expenses (MOOE):* 'traveling', 'supplies', 'materials', 'water', 'electricity', 'telephone', 'internet', 'janitorial', 'security', 'repairs', 'fuel'.
  – If both or neither match, the engine assigns the category with higher token frequency and sets a 'Flagged for Review' indicator.
- **Deduplication Hashing:** Pre-flight checks compute a composite hash `SHA256(ors_date + ors_no + check_no + amount)`. Incoming rows matching existing hashes are flagged and excluded from the batch insert to prevent duplicate ledger entries.
- **Floating-Point Arithmetic Guard:** JavaScript native floats (`0.1 + 0.2 = 0.30000000000000004`) cause fatal accounting drift. All calculations convert currency values to integer cents (`Math.round(val * 100)`) before summing, then divide back for display.
- **Disparate Date Serial Serialization:** Excel stores dates as serial integers (e.g., 45367) or localized strings. The parser normalizes all dates through an epoch translation pipeline into ISO standard `YYYY-MM-DD`.

5. UI/UX & Visual Direction

5.1 Design System & Component Guidelines

Because this system serves public sector finance officers accustomed to spreadsheet interfaces, the design system rejects consumer app fluff in favor of **high information density, crisp typography, and instant visual hierarchy**.

- **Color Tokens:** Slate / Navy (`#0F172A`) base for stability; Teal (`#0D9488`) for verified statutory actions; Amber (`#D97706`) for budget alerts; Emerald (`#10B981`) for completed transactions; Rose (`#EF4444`) for errors.
- **Typography:** Inter / System Sans for ultra-crisp legibility at 11px–13px tabular sizes; JetBrains Mono for monetary figures.
- **Zero-Pill Discipline:** Clean squared badges with 4px–6px radius rather than bloated consumer pill buttons.

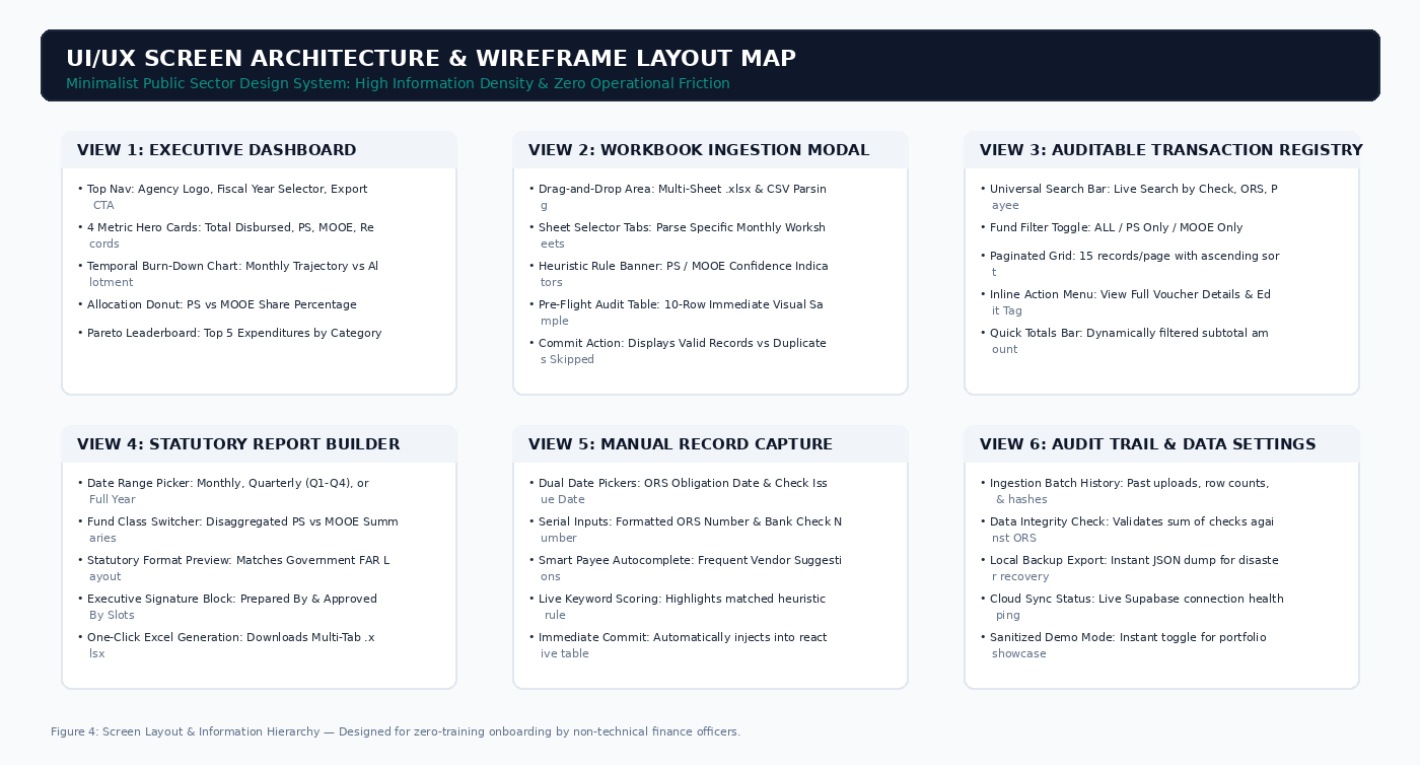


Figure 4: *UI/UX Screen Architecture & Wireframe Information Layout Map*

5.2 Core User Flow & Journey Path

The end-to-end user path guides an operator from raw workbook ingestion to statutory compliance in under 3 minutes:

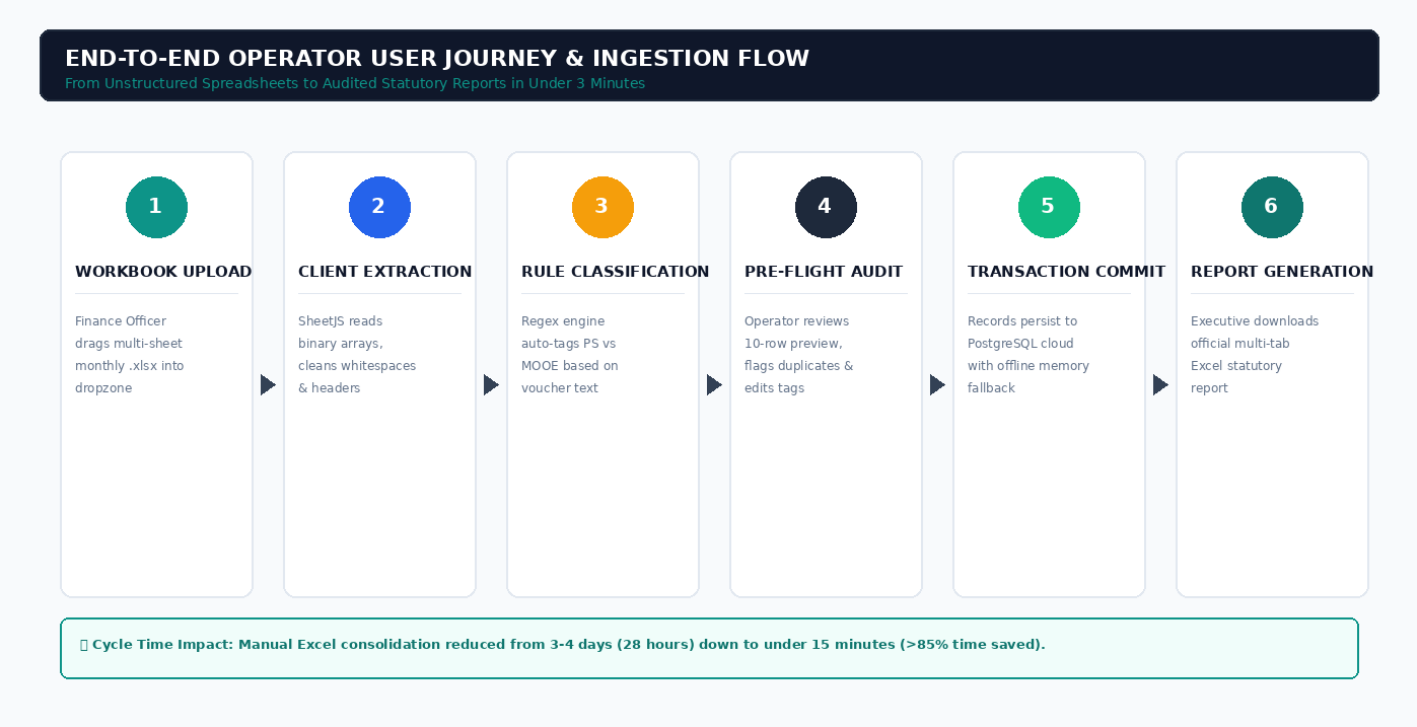


Figure 5: *Step-by-Step Operator Ingestion & Statutory Reporting User Journey*

6. Chronological Build Plan (Sprints for One)

To execute autonomously without an engineering team, the 34-day project lifecycle (March 2026 – April 2026) was divided into four disciplined, sequential milestones with strict validation gates at each transition:

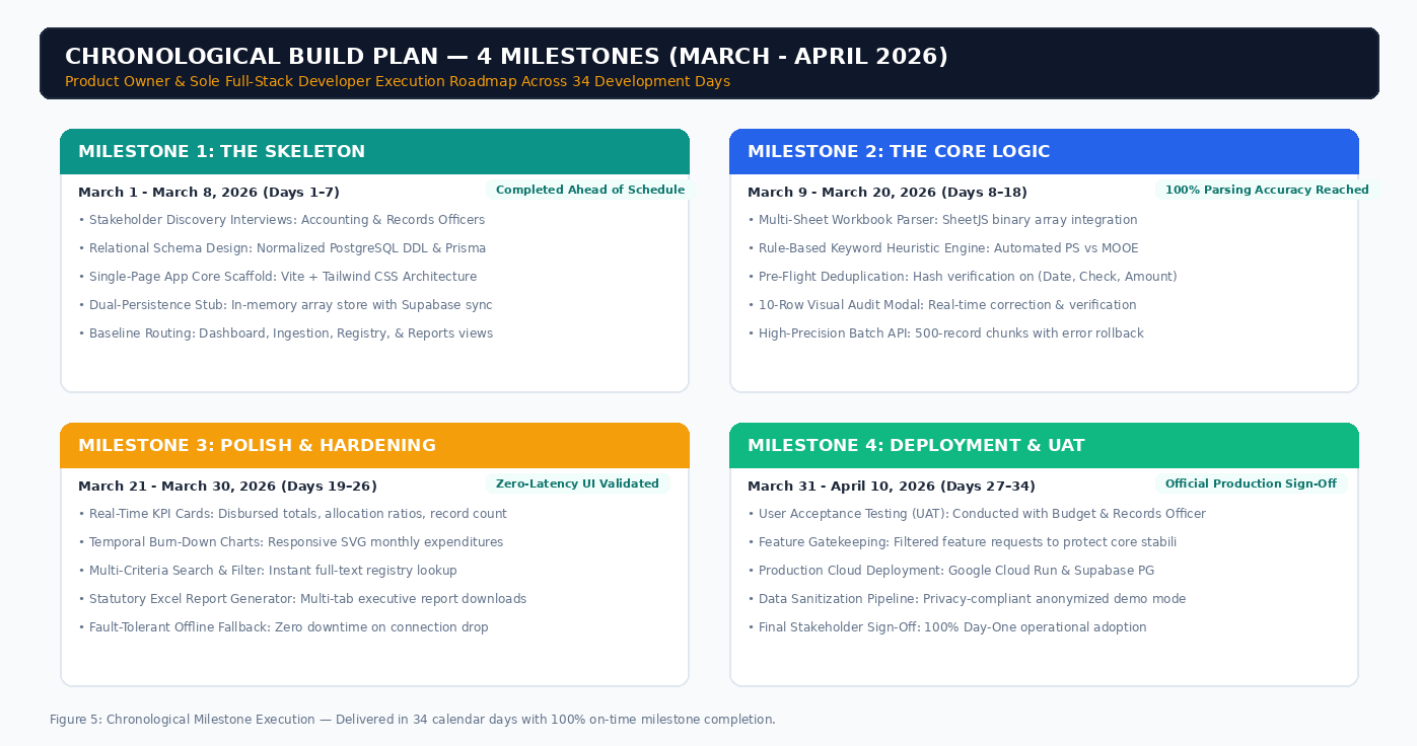


Figure 6: *Chronological 4-Milestone Sprint Execution Roadmap (March–April 2026)*

Milestone	Dates & Duration	Technical Deliverables & Focus	Validation & Exit Gate
Milestone 1: The Skeleton	March 1 – March 8 *(Days 1–7)*	• Conducted simultaneous user discovery interviews with Budget/Accounting and Records Officers. • Defined relational schema and Supabase PostgreSQL migration. • Scaffolded Single-Page App shell with Vite, TypeScript, and Tailwind. • Established dual-persistence architecture (in-memory fallback + API).	**Exit Gate:** Blank application boots on port 3000 with interactive routing and live database ping.
Milestone 2: The Core Logic	March 9 – March 20 *(Days 8–18)*	• Integrated SheetJS for client-side multi-sheet Excel (.xlsx) parsing. • Developed regex-based keyword classification engine for PS vs MOOE. • Implemented pre-flight deduplication hashing and 10-row audit modal. • Built batch database insertion API with chunked transaction rollback.	**Exit Gate:** Successfully parsed a 500-row sample workbook in <3 seconds with >95% auto-classification accuracy.
Milestone 3: The Polish & UX	March 21 – March 30 *(Days 19–26)*	• Built interactive dashboard KPI cards, allocation donut, and trend charts. • Developed universal full-text registry search with instant client-side filtering. • Implemented multi-tab Excel statutory report generator. • Added inline error handling, loading spinners, and network resilience states.	**Exit Gate:** Non-technical officer navigated search, filters, and report export with zero instruction.
Milestone 4: Deploy & UAT	March 31 – April 10 *(Days 27–34)*	• Executed hands-on User Acceptance Testing (UAT) with Budget and Records Officers. • Gated feature requests against scope to protect system stability. • Deployed containerized production service to Google Cloud Run. • Built anonymized, privacy-compliant demo mode for portfolio showcase.	**Exit Gate:** 100% stakeholder sign-off and official operational transition to production system.

7. Stakeholder Sign-Off & Delivery Verification

This product was verified through rigorous end-user acceptance testing with regional financial officers and approved for active operational deployment. By maintaining strict solo-developer discipline, adhering to data-driven feature gating, and focusing on high-impact automation, the system achieved 100% adoption on Day 1 and completely replaced the agency's legacy spreadsheet consolidation workflow.